

Studying Aberrations from the Fundamentals

1 Conventions and the mathematical tools

1.1 What is being described?

A ray describes a direction and trajectory in geometrical optics. An eikonal describes phase as a function of position. A surface sag describes the physical shape of a lens or mirror. These are distinct quantities even when some of them are measured in the same length units.

Symbol	Meaning	Units
$\lambda_0, k_0 = 2\pi/\lambda_0$	Vacuum wavelength and vacuum wavenumber	length; inverse length
n, n'	Refractive indices; a prime labels the output medium	dimensionless
ℓ	Geometrical arc length along a ray	length
$\mathcal{L} = \int n d\ell$	Optical path length	length
$S, W = S_a - S_0$	Eikonal; actual-minus-ideal wavefront error	length
$\mathbf{r} = (x, y, z)$	Three-dimensional position	length
$r = \sqrt{x^2 + y^2}$	Radial pupil coordinate when the origin is on axis	length
$a, \rho = r/a$	Physical pupil radius; normalized pupil radius	length; dimensionless
$u = x/a, v = y/a$	Normalized Cartesian pupil coordinates	dimensionless
$\mathbf{d} = d\mathbf{r}/d\ell$	Unit ray direction	dimensionless
$\mathbf{q} = (d_x, d_y)$	Transverse direction cosines, not pupil position	dimensionless
$\mathbf{s} = (dx/dz, dy/dz)$	Transverse ray slope	dimensionless
L	Axial separation of comparison and image planes	length
f	Focal length; it is not automatically equal to L	length

Unless a mirror example says otherwise, propagation is toward $+z$. For transmitted systems a spherical radius is positive when its centre of curvature lies downstream of the vertex. We use time dependence $e^{-i\omega t}$ and real positive refractive indices. An *exact* ray formula is exact within geometrical optics, not an exact solution of the electromagnetic diffraction problem.

1.2 Units, derivatives, and coordinate normalization

The conversions

$$1 \text{ mm} = 10^{-3} \text{ m}, \quad 1 \mu\text{m} = 10^{-3} \text{ mm}, \quad 1 \text{ nm} = 10^{-6} \text{ mm}$$

must be applied before combining wavelength, optical path, and geometry. For example, $633 \text{ nm} = 0.000633 \text{ mm}$. A path error W produces phase error

$$\Delta\phi = k_0 W = \frac{2\pi W}{\lambda_0}. \quad (1.1)$$

Thus $W = 31.65 \text{ nm}$ is 0.05 waves at 633 nm, giving $\Delta\phi = 0.31416$ radians. Angles in Taylor series are in radians: $1^\circ = \pi/180$ radians.

For a sag $h(r) = cr^2/2$, the definition of derivative gives

$$\frac{h(r + \Delta r) - h(r)}{\Delta r} = cr + \frac{c\Delta r}{2} \rightarrow cr.$$

Hence $h'(r) = cr$ and $h''(r) = c$. A partial derivative varies one coordinate while holding the others fixed. For example,

$$W = A(x^2 + y^2)^2 \implies \nabla_{\perp} W = 4A(x^2 + y^2)(x, y). \quad (1.2)$$

If W has units of length, A here has units of inverse length cubed. Writing the same function as $W = B\rho^4$ instead gives $B = Aa^4$, which has units of length.

Define $\widetilde{W}(u, v) = W(au, av)$. The chain rule gives

$$\frac{\partial W}{\partial x} = \frac{1}{a} \frac{\partial \widetilde{W}}{\partial u}, \quad \frac{\partial W}{\partial y} = \frac{1}{a} \frac{\partial \widetilde{W}}{\partial v}, \quad \nabla_{\perp} W = \frac{1}{a} \nabla_{u,v} \widetilde{W}. \quad (1.3)$$

The normalized coordinates are dimensionless; $\widetilde{W}$ remains an optical path in length units.

1.3 Disk averages and wavefront statistics

For $x = a\rho \cos \theta$, $y = a\rho \sin \theta$, a polar area element has radial width $d\rho$ and tangential width $r d\theta$. Therefore

$$dA = r d\rho d\theta = a^2 \rho d\rho d\theta, \quad A_{\text{pupil}} = \int_0^{2\pi} \int_0^1 a^2 \rho d\rho d\theta = \pi a^2.$$

Uniform full-disk area averaging means

$$\langle F \rangle = \frac{1}{\pi} \int_0^{2\pi} \int_0^1 F(\rho, \theta) \rho d\rho d\theta. \quad (1.4)$$

For $m > -2$,

$$\langle \rho^m \rangle = 2 \int_0^1 \rho^{m+1} d\rho = \frac{2}{m+2}. \quad (1.5)$$

In particular, $\langle \rho^2 \rangle = 1/2$, $\langle \rho^4 \rangle = 1/3$, $\langle \rho^6 \rangle = 1/4$, and $\langle \rho^8 \rangle = 1/5$.

Expanding the square gives the piston-removed variance

$$\sigma_W^2 = \langle (W - \langle W \rangle)^2 \rangle = \langle W^2 \rangle - \langle W \rangle^2. \quad (1.6)$$

For $W = B\rho^2$,

$$\langle W \rangle = B/2, \quad \sigma_W^2 = B^2 \left(\frac{1}{3} - \frac{1}{4} \right) = \frac{B^2}{12}, \quad \sigma_W = \frac{|B|}{\sqrt{12}}.$$

For comparison, $W = B\rho^4$ gives $\sigma_W^2 = B^2(1/5 - 1/9) = 4B^2/45$. Both raw monomials have peak-to-valley magnitude $|B|$, but different RMS values. The weight ρ is essential; uniform weighting in radius is a different measurement from uniform weighting in pupil area.

1.4 Taylor series, distance expansions, and matrix order

Repeated differentiation of $f(\epsilon) = (1 + \epsilon)^p$ gives

$$f^{(k)}(0) = p(p-1) \cdots (p-k+1).$$

Inserting these derivatives into Taylor's formula yields

$$(1 + \epsilon)^p = 1 + p\epsilon + \frac{p(p-1)}{2}\epsilon^2 + \frac{p(p-1)(p-2)}{6}\epsilon^3 + \dots$$

The cases needed below include

$$\sqrt{1 + \epsilon} = 1 + \frac{\epsilon}{2} - \frac{\epsilon^2}{8} + \frac{\epsilon^3}{16} - \frac{5\epsilon^4}{128} + O(\epsilon^5), \quad (1.7)$$

$$\sqrt{1 - \epsilon} = 1 - \frac{\epsilon}{2} - \frac{\epsilon^2}{8} - \frac{\epsilon^3}{16} - \frac{5\epsilon^4}{128} + O(\epsilon^5), \quad (1.8)$$

$$(1 + \epsilon)^{-1} = 1 - \epsilon + \epsilon^2 + O(\epsilon^3). \quad (1.9)$$

An $O(\epsilon^k)$ remainder is bounded in magnitude by a constant times $|\epsilon|^k$ sufficiently near zero. It is not a global accuracy promise.

For $L > 0$,

$$\sqrt{L^2 + r^2} = L + \frac{r^2}{2L} - \frac{r^4}{8L^3} + \frac{r^6}{16L^5} + O(r^8/L^7). \quad (1.10)$$

For a sphere opening toward $+z$, with positive radius magnitude R ,

$$h(r) = R - \sqrt{R^2 - r^2} = \frac{r^2}{2R} + \frac{r^4}{8R^3} + \frac{r^6}{16R^5} + \frac{5r^8}{128R^7} + \dots \quad (1.11)$$

The expansion parameters are r^2/L^2 and r^2/R^2 , respectively.

For vectors, $\mathbf{a} \cdot \mathbf{b} = \sum_i a_i b_i$ and $|\mathbf{a}|^2 = \mathbf{a} \cdot \mathbf{a}$. A straight ray with unit direction $\mathbf{d}$ is $\mathbf{r}(\ell) = \mathbf{r}_0 + \ell \mathbf{d}$. For matrices, if M_1 acts first and M_2 second, the result is $M_2 M_1 \mathbf{v}$. This ordering matters when a translation and refraction are composed.

2 Optical path, phase, and the eikonal

2.1 From wavelength to optical path

In a homogeneous transparent medium, the phase wavelength is $\lambda = \lambda_0/n$. Travelling a geometrical distance $d\ell$ adds

$$d\phi = \frac{2\pi}{\lambda} d\ell = k_0 n d\ell.$$

Summing along a ray gives

$$\mathcal{L} = \int n d\ell, \quad \phi = \phi_{\text{initial}} + k_0 \mathcal{L}. \quad (2.1)$$

For piecewise homogeneous media, $\mathcal{L} = \sum_j n_j \ell_j$. Thus 10 mm of air followed by 5 mm of index 1.5 has geometrical length 15 mm and optical path 17.5 mm.

2.2 Eikonal means phase expressed as a length

Write a monochromatic scalar field as

$$U(\mathbf{r}, t) = A(\mathbf{r}) e^{i[k_0 S(\mathbf{r}) - \omega t]}. \quad (2.2)$$

The spatial phase is $k_0 S$, so S is phase measured in optical-length units. Along a ray, (2.1) gives

$$S(\mathbf{r}_2) - S(\mathbf{r}_1) = \int_{r_1}^{r_2} n d\ell. \quad (2.3)$$

The integral follows the ray branch being described; its constant of integration fixes the phase reference.

A wavefront is a surface $S(\mathbf{r}) = \text{constant}$ at fixed time. If $\mathbf{t}$ is tangent to that surface, differentiation along it gives $\nabla S \cdot \mathbf{t} = 0$. Therefore ∇S is normal to the wavefront.

2.3 Derive the eikonal equation from the scalar wave equation

For the scalar approximation in an isotropic medium,

$$\nabla^2 U + k_0^2 n^2 U = 0.$$

Substitute $U = Ae^{ik_0 S}$. The first derivative is

$$\nabla U = e^{ik_0 S} (\nabla A + ik_0 A \nabla S).$$

Taking the divergence and using the product rule,

$$\nabla^2 U = e^{ik_0 S} [\nabla^2 A + 2ik_0 \nabla A \cdot \nabla S + ik_0 A \nabla^2 S - k_0^2 A |\nabla S|^2]. \quad (2.4)$$

After substitution and cancellation of the exponential,

$$\nabla^2 A + 2ik_0 \nabla A \cdot \nabla S + ik_0 A \nabla^2 S + k_0^2 A (n^2 - |\nabla S|^2) = 0. \quad (2.5)$$

When wavelength is short compared with the amplitude and material variation scales, the leading k_0^2 terms must cancel:

$$\boxed{|\nabla S|^2 = n^2.} \quad (2.6)$$

Locally the phase is $k_0 S(\mathbf{r}_0) + k_0 \nabla S(\mathbf{r}_0) \cdot \Delta \mathbf{r}$. It is a local plane wave with wavevector $\mathbf{k} = k_0 \nabla S$. In an isotropic medium its ray direction is along $\mathbf{k}$, giving

$$\boxed{\mathbf{d} = \frac{\nabla S}{n}.} \quad (2.7)$$

This reproduces textbook Eq. (2.6). The slowly varying amplitude and single-ray-branch assumptions need care at caustics, near sharp boundaries, and where diffraction is essential.

2.4 Why equal phase at a target is the ideal focusing condition

At pupil point Q , the initial phase is $k_0 S(Q)$. The scalar geometrical propagation phase to a target F adds $k_0 n' |F - Q|$. For contributions with amplitudes $a_j \geq 0$, the field at F has the form

$$U(F) = \sum_j a_j e^{ik_0 [S(Q_j) + n' |F - Q_j|]}.$$

For two terms,

$$|a_1 e^{i\phi_1} + a_2 e^{i\phi_2}|^2 = a_1^2 + a_2^2 + 2a_1 a_2 \cos(\phi_1 - \phi_2).$$

For fixed amplitudes this is largest when their phases agree modulo 2π . For the whole pupil, $|U(F)| \leq \sum_j a_j$, with equality when all nonzero phasors align. The ideal smooth coherent reference therefore obeys

$$\boxed{S_{\text{ideal}}(Q) + n' |F - Q| = C.} \quad (2.8)$$

Strictly, equality modulo λ_0 suffices for the optical lengths. On a connected pupil with a smooth unwrapped eikonal, the integer-wavelength offset is constant and can be absorbed into C .

There is also a ray proof. For $Q = (x, y, 0)$ and $F = (0, 0, L)$, the desired ray direction is $(-x, -y, L)/\sqrt{L^2 + r^2}$. Hence $S_x = -n'x/\sqrt{L^2 + r^2}$ and $S_y = -n'y/\sqrt{L^2 + r^2}$. Since the distance derivatives are $x/\sqrt{L^2 + r^2}$ and $y/\sqrt{L^2 + r^2}$, $\nabla_\perp (S + n'\sqrt{L^2 + r^2}) = 0$. This is exactly (2.8).

Interpretation. An outer pupil point has a longer remaining distance. An ideal focusing wave has a starting phase that compensates that difference. Different remaining distances do not by themselves imply aberration. An ideal focus is an unaberrated coherent reference; a finite aperture still produces a diffraction pattern of finite width.

3 Reflection and refraction from optical path

3.1 Fermat's principle and Snell's law

Fermat's principle states that the optical path is stationary under allowed small path variations with the endpoints fixed. Stationary means zero first variation; it does not necessarily mean a global minimum.

Put an interface at $z = 0$, a source at $A = (x_A, -h_1)$ in index n_1 , and a destination at $B = (x_B, h_2)$ in index n_2 . A trial crossing point is $P = (x, 0)$, so

$$\mathcal{L}(x) = n_1 \sqrt{(x - x_A)^2 + h_1^2} + n_2 \sqrt{(x_B - x)^2 + h_2^2}.$$

Differentiate each square root:

$$\frac{d\mathcal{L}}{dx} = n_1 \frac{x - x_A}{\sqrt{(x - x_A)^2 + h_1^2}} - n_2 \frac{x_B - x}{\sqrt{(x_B - x)^2 + h_2^2}}.$$

The ratios are the signed sines of the incident and transmitted angles measured from the normal. Stationarity therefore gives

$$\boxed{n_1 \sin i = n_2 \sin t.} \quad (3.1)$$

For reflection, both endpoints are in the same medium and the two path segments lie on the same side of the surface. The corresponding stationary condition preserves the tangential direction component while the normal component reverses.

3.2 Exact vector reflection

Let $\mathbf{N}$ be either unit normal at the reflecting point. Decompose an incident unit direction into

$$\mathbf{d}_n = (\mathbf{d}_{\text{in}} \cdot \mathbf{N})\mathbf{N}, \quad \mathbf{d}_t = \mathbf{d}_{\text{in}} - \mathbf{d}_n.$$

Specular reflection preserves $\mathbf{d}_t$ and reverses $\mathbf{d}_n$, hence

$$\boxed{\mathbf{d}_{\text{out}} = \mathbf{d}_{\text{in}} - 2(\mathbf{d}_{\text{in}} \cdot \mathbf{N})\mathbf{N}.} \quad (3.2)$$

Changing $\mathbf{N}$ to $-\mathbf{N}$ leaves this expression unchanged. Its norm is one because the orthogonal tangential and normal components retain their respective squared magnitudes. At normal incidence, $(0, 0, -1)$ becomes $(0, 0, 1)$.

3.3 Exact vector refraction and total internal reflection

Choose $\mathbf{N}$ to point into the incident medium, so $\mathbf{d}_{\text{in}} \cdot \mathbf{N} < 0$. Define

$$\eta = \frac{n_1}{n_2}, \quad c_i = -\mathbf{d}_{\text{in}} \cdot \mathbf{N}.$$

The incident tangential vector is $\mathbf{d}_{\text{in}} + c_i \mathbf{N}$. Snell's law scales it by η . Unit normalization determines the transmitted normal magnitude:

$$c_t^2 = 1 - \eta^2 |\mathbf{d}_{\text{in}} + c_i \mathbf{N}|^2 = 1 - \eta^2 (1 - c_i^2).$$

The transmitted ray points away from the incident medium, so its normal component is $-c_t \mathbf{N}$. Therefore

$$\boxed{c_t = \sqrt{1 - \eta^2 (1 - c_i^2)}, \quad \mathbf{d}_{\text{tr}} = \eta \mathbf{d}_{\text{in}} + (\eta c_i - c_t) \mathbf{N}.} \quad (3.3)$$

If the square root is imaginary, there is no propagating transmitted ray in this model. For $n_1 > n_2$, the threshold is

$$\sin i_c = \frac{n_2}{n_1}.$$

For $n_1 = 1.5$ and $n_2 = 1$, $i_c \simeq 41.81^\circ$. At normal incidence $\mathbf{d}_{\text{in}} = -\mathbf{N}$ and the formula returns $\mathbf{d}_{\text{tr}} = -\mathbf{N}$, as it should.

4 First-order imaging: surfaces, lenses, and mirrors

4.1 Paraxial power of a spherical refracting surface

In a meridional section, let the ray height be y and its signed angle to $+z$ be ϑ . Near the axis,

$$\sin \vartheta \simeq \vartheta, \quad \tan \vartheta \simeq \vartheta, \quad \cos \vartheta \simeq 1.$$

These approximations retain only first-degree terms in heights and angles.

A spherical surface of signed radius R has sag $h(y) \simeq y^2/(2R)$ and normal angle $\alpha_N \simeq -y/R$. Linearizing Snell's law about that normal gives

$$n(\vartheta - \alpha_N) = n'(\vartheta' - \alpha_N).$$

Rearranging,

$$\boxed{n'\vartheta' = n\vartheta - \frac{n' - n}{R}y.} \quad (4.1)$$

Define the surface power $\Phi = (n' - n)/R$ and reduced angle $p = n\vartheta$. The refraction law becomes $p' = p - \Phi y$.

For an on-axis real object a distance $s_o > 0$ before the surface and real image a distance $s_i > 0$ after it, $\vartheta \simeq y/s_o$ and $\vartheta' \simeq -y/s_i$. Substitution gives

$$\boxed{\frac{n}{s_o} + \frac{n'}{s_i} = \frac{n' - n}{R}.} \quad (4.2)$$

The real conjugate distances use the indicated positive convention; virtual conjugates acquire negative values in this formula.

4.2 Thin-lens equation from rays and from optical path

For a thin lens of glass index n_g surrounded by air, add the powers of its two surfaces:

$$\Phi = (n_g - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right), \quad \vartheta_{\text{out}} = \vartheta_{\text{in}} - \Phi y. \quad (4.3)$$

For collimated input $\vartheta_{\text{in}} = 0$, the ray reaches the axis after distance f when $y + f(-\Phi y) = 0$. Thus

$$\boxed{\frac{1}{f} = (n_g - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right).} \quad (4.4)$$

For finite conjugates, $\vartheta_{\text{in}} = y/s_o$ and $\vartheta_{\text{out}} = -y/s_i$, giving

$$\boxed{\frac{1}{s_o} + \frac{1}{s_i} = \frac{1}{f}, \quad m = \frac{y_i}{y_o} = -\frac{s_i}{s_o}.} \quad (4.5)$$

The magnification follows by tracing the central paraxial ray.

For an independent phase derivation, the change in biconvex thickness is

$$t(y) - t(0) \simeq \frac{y^2}{2} \left(\frac{1}{R_2} - \frac{1}{R_1} \right).$$

Between fixed planes, replacing air by glass adds $(n_g - 1)[t(y) - t(0)] = -y^2/(2f)$. The paths from the object and to the image add $y^2/(2s_o)$ and $y^2/(2s_i)$ to quadratic order. Equal phase requires

$$\frac{y^2}{2} \left(\frac{1}{s_o} + \frac{1}{s_i} - \frac{1}{f} \right) = 0,$$

reproducing the ray result.

For $n_g = 1.5$, $R_1 = 50$ mm, and $R_2 = -50$ mm, $f = 50$ mm. An object at $s_o = 200$ mm has $s_i = 66.6667$ mm and $m = -1/3$.

Keep distinct. The thickness-only calculation is sufficient here at Gaussian order. Extending it to high powers does not automatically give the exact aberration of a real lens: internal ray bending, oblique glass paths, and different intersection heights also contribute.

4.3 Paraxial focal length and imaging law of a spherical mirror

For this mirror only, use incident light along $-z$ and reflected light toward $+z$. The surface opens toward $+z$:

$$h(r) = R - \sqrt{R^2 - r^2} \simeq \frac{r^2}{2R}.$$

In a meridional section its unit normal is approximately $\mathbf{N} = (-r/R, 0, 1)$ to first order. Applying (3.2) to $\mathbf{d}_{\text{in}} = (0, 0, -1)$ gives $\mathbf{d}_{\text{out}} \simeq (-2r/R, 0, 1)$. The ray height after axial distance f is approximately $r - 2fr/R$, so

$$\boxed{f = \frac{R}{2}}. \quad (4.6)$$

For finite positive object and image distances s_o, s_i in front of this mirror, the geometrical path via radius r is

$$\sqrt{r^2 + [s_o - h(r)]^2} + \sqrt{r^2 + [s_i - h(r)]^2}.$$

For either conjugate s ,

$$\sqrt{r^2 + [s - h(r)]^2} = s - h(r) + \frac{r^2}{2s} + O(r^4) = s + r^2 \left(\frac{1}{2s} - \frac{1}{2R} \right) + O(r^4).$$

Summing and setting the quadratic coefficient to zero gives

$$\boxed{\frac{1}{s_o} + \frac{1}{s_i} = \frac{2}{R} = \frac{1}{f}}. \quad (4.7)$$

The omitted terms generally prevent a sphere from focusing all finite-aperture rays at one point. The paraboloid at the end provides an exact on-axis counterexample: it focuses an axial plane wave without geometrical spherical aberration.

4.4 Matrices, thick lenses, principal planes, and invariants

Use the ray vector $(y, p)^T$ with $p = n\vartheta$. Translation by distance d in index n gives $y' = y + (d/n)p$ and $p' = p$. A refracting surface preserves height and changes p to $p - \Phi y$. Their matrices are

$$T(d, n) = \begin{pmatrix} 1 & d/n \\ 0 & 1 \end{pmatrix}, \quad R(\Phi) = \begin{pmatrix} 1 & 0 \\ -\Phi & 1 \end{pmatrix}. \quad (4.8)$$

For a lens of thickness d and index n_g ,

$$M = R(\Phi_2)T(d, n_g)R(\Phi_1) \quad (4.9)$$

$$= R(\Phi_2) \begin{pmatrix} 1 - d\Phi_1/n_g & d/n_g \\ -\Phi_1 & 1 \end{pmatrix} \quad (4.10)$$

$$= \begin{pmatrix} 1 - d\Phi_1/n_g & d/n_g \\ -\Phi_1 - \Phi_2 + d\Phi_1\Phi_2/n_g & 1 - d\Phi_2/n_g \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix}. \quad (4.11)$$

The effective power is $\Phi_{\text{eff}} = -C$. For air on both sides, the effective focal length is $f_{\text{EFL}} = -1/C$. A collimated ray $(y_0, 0)$ exits as (Ay_0, Cy_0) . After a further air distance b , its height is $(A + bC)y_0$, giving

$$\boxed{f_{\text{EFL}} = -\frac{1}{C}, \quad \text{BFL} = -\frac{A}{C}.} \quad (4.12)$$

BFL is measured from the rear vertex. EFL is measured from the rear principal plane, the effective reference plane for first-order focusing.

For the preceding biconvex lens with $d = 5 \text{ mm}$, $\Phi_1 = \Phi_2 = 0.01 \text{ mm}^{-1}$ and

$$A = D = \frac{29}{30}, \quad B = \frac{10}{3} \text{ mm}, \quad C = -\frac{59}{3000} \text{ mm}^{-1}.$$

Thus $f_{\text{EFL}} = 3000/59 = 50.84746 \text{ mm}$ and $\text{BFL} = 2900/59 = 49.15254 \text{ mm}$. If the first vertex is $z = 0$, the paraxial image of an axial plane wave is at $z = 5 + \text{BFL} = 54.15254 \text{ mm}$.

Include the object-to-system and system-to-image translations in a total matrix. Then $y_i = Ay_o + Bp_o$. All rays from the same object point have fixed y_o and variable p_o , so the imaging condition is

$$\boxed{B = 0, \quad y_i = Ay_o.} \quad (4.13)$$

Both elementary matrices have determinant one, giving $AD - BC = 1$. For two rays,

$$y'_1 p'_2 - y'_2 p'_1 = (Ay_1 + Bp_1)(Cy_2 + Dp_2) - (Ay_2 + Bp_2)(Cy_1 + Dp_1) \quad (4.14)$$

$$= (AD - BC)(y_1 p_2 - y_2 p_1) = y_1 p_2 - y_2 p_1. \quad (4.15)$$

This preserved combination is a useful first-order consistency check.

5 Pupils and reference wavefronts

5.1 Pupil position selects a ray; field position selects an object point

For a fixed object point, the aperture stop limits the admitted ray cone. It can be an iris, lens rim, or mirror boundary. The entrance pupil is its image through the preceding optics, seen from object space; the exit pupil is its image through the following optics, seen from image space. Either pupil can be virtual and neither must coincide with a physical lens surface.

A chief ray passes through the stop centre. A marginal ray passes through its edge. A field stop limits admitted object positions. Vignetting clips some field bundles and changes the pupil shape or weighting.

Change	What it selects
Change pupil coordinate	A different ray from the same object point
Change field coordinate	A different object point or incident field angle
Change reference image	A different target against which those rays are judged

For an object at infinity in air, the f-number is $N = f_{\text{EFL}}/D_{\text{entrance}}$. The image-space numerical aperture is $\text{NA} = n' \sin \alpha$. If the relevant pupil radius is a , its axial distance to focus is L , and the cone is paraxial, $\tan \alpha = a/L$ gives $\text{NA} \simeq n'a/L$. When additionally $L \simeq f_{\text{EFL}}$, the relevant diameter is $2a = D_{\text{entrance}}$, and $n' = 1$, this becomes $\text{NA} \simeq 1/(2N)$. The shortcut is not general at finite conjugates or large aperture.

5.2 Specify the ideal before defining the error

Let the comparison plane be $z = z_e$ and the reference image be $F = (X_0, Y_0, z_f)$, with $L = z_f - z_e > 0$. Equation (2.8) gives textbook Eq. (4.1):

$$S_0(x, y) = C - n' \sqrt{(x - X_0)^2 + (y - Y_0)^2 + L^2}. \quad (5.1)$$

The sign is fixed by convergence:

$$\frac{\partial S_0}{\partial x} = - \frac{n'(x - X_0)}{\sqrt{(x - X_0)^2 + (y - Y_0)^2 + L^2}},$$

whose direction component points toward X_0 . The wavefront error is textbook Eq. (4.2):

$$\boxed{W(x, y) = S_a(x, y) - S_0(x, y)}. \quad (5.2)$$

Equivalently, $W(Q) = S_a(Q) + n'|F - Q| - C$. The segment from Q to F defines the subtracted ideal. An aberrated physical ray is not thereby forced to pass through F .

Interpretation. An ideal wave can have nonzero slope and curvature while $W = 0$. At an off-centre pupil point its ray is inclined toward the image, but it has no directional error relative to the ideal ray at that same point. The word error describes a mismatch with the specified reference.

5.3 Piston: constant phase across the pupil

For $W = P$, the phase multiplier is $e^{ik_0 P}$. Every contribution to a coherent image receives the same factor, so

$$U_{\text{new}} = e^{ik_0 P} U_{\text{old}}, \quad |U_{\text{new}}|^2 = |U_{\text{old}}|^2, \quad \nabla_{\perp} P = 0.$$

Piston changes neither ray directions nor the intensity of that isolated coherent image. Differential piston between coherent interferometer arms or mirror segments can change interference.

With a specified pupil weight, mean-piston removal is

$$P_{\text{mean}} = \langle W \rangle, \quad W_{\text{residual}} = W - \langle W \rangle.$$

Setting $W(0) = 0$ is a different reference choice. For example, $W = B\rho^2$ has zero centre value but nonzero mean $B/2$. On the uniformly weighted full disk, mean removal gives $B(\rho^2 - 1/2)$. A constant Taylor coefficient and the fitted full-pupil piston need not be numerically equal.

5.4 Where linear terms come from: tilt relative to the reference

A smooth pupil function can be Taylor-expanded:

$$W(x, y) = W(0) + W_x(0)x + W_y(0)y \quad (5.3)$$

$$+ \frac{1}{2} \left[W_{xx}(0)x^2 + 2W_{xy}(0)xy + W_{yy}(0)y^2 \right] + \dots \quad (5.4)$$

The linear terms are present when the reference and actual directions differ at the expansion point; they are not created by merely choosing an off-centre pupil point.

For a plane wave in homogeneous index n' , $\mathbf{k} = n'k_0\mathbf{d}$ and $U = Ae^{i(\mathbf{k}\cdot\mathbf{r}+\phi_a)}$. Comparing with Ae^{ik_0S} gives

$$S_a(\mathbf{r}) = n'\mathbf{d} \cdot \mathbf{r} + C_a.$$

On $z = 0$ this is $n'd_x x + n'd_y y + C_a$. The axial reference plane wave has $S_0 = n'z + C_r$, which is constant only across a fixed z plane. Subtracting on $z = 0$ yields

$$\boxed{W = P + b_x x + b_y y, \quad b_x = n'd_x, \quad b_y = n'd_y.} \quad (5.5)$$

For an inclination α in the xz plane, $W = P + n'x \sin \alpha \simeq P + n'\alpha x$. Its gradient is constant. Section 6 will show that this produces a uniform image shift paraxially.

5.5 Quadratic radial error and a shifted reference focus

For an axial focus, use (1.10) in (5.1):

$$S_0(r; L) = P(L) - \frac{n'r^2}{2L} + \frac{n'r^4}{8L^3} - \frac{n'r^6}{16L^5} + \dots \quad (5.6)$$

The quadratic term in S_0 is required for ideal convergence. A radial quadratic term in the *difference* W is defocus.

Hold S_a fixed on the same comparison plane and replace the target distance L by $L + \Delta L$. Then

$$W_{\text{new}} - W_{\text{old}} = S_0(r; L) - S_0(r; L + \Delta L) \quad (5.7)$$

$$= \Delta P + \frac{n'r^2}{2} \left(\frac{1}{L + \Delta L} - \frac{1}{L} \right) - \frac{n'r^4}{8} \left(\frac{1}{(L + \Delta L)^3} - \frac{1}{L^3} \right) + \dots \quad (5.8)$$

The reciprocal difference is $-\Delta L/[L(L + \Delta L)] \simeq -\Delta L/L^2$ when $|\Delta L| \ll L$. Therefore textbook Eq. (4.3) is

$$\boxed{W_{\text{new}} = W_{\text{old}} - \frac{n'\Delta L}{2L^2} r^2 + \text{piston} + \text{higher-order terms.}} \quad (5.9)$$

The next term, first order in ΔL , is $3n'\Delta L r^4/(8L^4)$. Both a small pupil angle r/L and a small relative shift $\Delta L/L$ have been used.

For an exact check, choose the on-axis reference values of both ideal eikonals to be zero. Then

$$W_{\text{new}} - W_{\text{old}} = n' \left[\sqrt{(L + \Delta L)^2 + r^2} - \sqrt{L^2 + r^2} - \Delta L \right].$$

Differentiating at $\Delta L = 0$ gives $n'[L/\sqrt{L^2 + r^2} - 1]\Delta L$. Its small- r/L expansion reproduces the negative quadratic coefficient.

For $\Delta L > 0$, the new ideal wave converges less strongly and its quadratic eikonal is less negative. Subtracting it makes the error more negative away from the centre. If the unchanged actual rays focus at L , a ray starting at x follows $x(z) = x(1 - z/L)$ and reaches the new plane at

$x(L + \Delta L) = -x\Delta L/L$. This is defocus at the displaced plane, even though the actual focus has not moved.

With $\rho = r/a$ and $W = B\rho^2$, (5.9) gives

$$\boxed{\Delta B = -\frac{n'a^2\Delta L}{2L^2}}. \quad (5.10)$$

For $n' = 1$, $a = 10$ mm, $L = 200$ mm, and $\Delta L = 0.1$ mm, $\Delta B = -0.000125$ mm = -125 nm. This is the change in edge-minus-centre quadratic OPD.

5.6 Reference changes, physical changes, and general quadratic terms

A reference change gives $\Delta W = -\Delta S_0$ at fixed actual wave. A physical change gives $\Delta W = +\Delta S_a$ at fixed reference. For example, adding gr^2 to the physical eikonal adds phase k_0gr^2 and changes the actual convergence according to

$$-\frac{n'r^2}{2L} + gr^2 = -\frac{n'r^2}{2} \left(\frac{1}{L} - \frac{2g}{n'} \right).$$

As long as it remains converging, its new paraxial focus satisfies $1/L_{a,\text{new}} = 1/L - 2g/n'$. A small positive g moves that physical focus farther away.

For an actual perfect focus at L_a and reference at L ,

$$W(r) \simeq P + \frac{n'}{2} \left(\frac{1}{L} - \frac{1}{L_a} \right) r^2.$$

It vanishes apart from piston when $L_a = L$. Finally, an arbitrary quadratic form is

$$Ax^2 + 2Bxy + Cy^2 = \frac{A+C}{2}(x^2 + y^2) + \frac{A-C}{2}(x^2 - y^2) + 2Bxy.$$

Only the rotationally symmetric first term is pure radial defocus. The remaining terms represent unequal directional curvature. Linear/quadratic here refers to pupil powers, not to the separate Seidel order convention.

6 From wavefront slope to ray error: complete Section 4.4

6.1 Specify the two rays and the error being measured

Set the output comparison plane to $z = 0$ and the image plane to $z = L$. Write $\boldsymbol{\xi} = (x, y)$ for a physical transverse position on the output plane. The actual and ideal rays start at this same point and have unit directions $\boldsymbol{d}$ and $\boldsymbol{d}_0$. Define the transverse ray-intercept error

$$\Delta \boldsymbol{\xi} = \boldsymbol{\xi}_{\text{image},a} - \boldsymbol{\xi}_{\text{image},0}. \quad (6.1)$$

It is measured across the image plane, in length units. A uniform displacement shifts a point image; varying displacements across the pupil produce a geometrical blur. No numerical mistake is implied by the word error.

6.2 Wavefront gradient gives a direction-cosine difference

Apply (2.7) to each wave in the same output index n' :

$$\boldsymbol{q} = \frac{\nabla_{\perp} S_a}{n'}, \quad \boldsymbol{q}_0 = \frac{\nabla_{\perp} S_0}{n'}.$$

Subtracting and using (5.2),

$$\delta \mathbf{q} = \mathbf{q} - \mathbf{q}_0 = \frac{\nabla_{\perp} W}{n'}. \quad (6.2)$$

This is an exact difference within the eikonal model on the stated plane. Here $\delta \mathbf{q}$ denotes the finite difference; it need not yet be linearized.

6.3 Direction and slope use different independent distances

Since ℓ is geometrical arc length, the full unit direction is

$$\mathbf{d} = \frac{d\mathbf{r}}{d\ell}, \quad \mathbf{q} = \left(\frac{dx}{d\ell}, \frac{dy}{d\ell} \right).$$

The axial coordinate z is generally not the distance travelled along the inclined ray. By the chain rule,

$$\frac{dx}{dz} = \frac{dx/d\ell}{dz/d\ell} = \frac{d_x}{d_z}, \quad \frac{dy}{dz} = \frac{d_y}{d_z}.$$

For the forward branch, $d_z = \sqrt{1 - |\mathbf{q}|^2} > 0$. Therefore

$$s(\mathbf{q}) = \frac{\mathbf{q}}{\sqrt{1 - |\mathbf{q}|^2}}. \quad (6.3)$$

In a meridional plane, $q_x = \sin \alpha$, $d_z = \cos \alpha$, and $s_x = \tan \alpha$. The slope is dimensionless; at finite angle it is not the angle in radians.

6.4 Exact propagation and textbook Eq. (4.5)

The actual ray is $\mathbf{R}(\ell) = (x, y, 0) + \ell \mathbf{d}$. At the image plane, $\ell d_z = L$, giving

$$\boldsymbol{\xi}_{\text{image,a}} = \boldsymbol{\xi} + L \frac{\mathbf{q}}{d_z} = \boldsymbol{\xi} + L s(\mathbf{q}).$$

The ideal ray similarly gives $\boldsymbol{\xi}_{\text{image,0}} = \boldsymbol{\xi} + L s(\mathbf{q}_0)$. Subtracting cancels the common starting position:

$$\Delta \boldsymbol{\xi} = L \left[s \left(\mathbf{q}_0 + \frac{\nabla_{\perp} W}{n'} \right) - s(\mathbf{q}_0) \right]. \quad (6.4)$$

This is textbook Eq. (4.5). The two directions must both be forward propagating, so $|\mathbf{q}_0| < 1$ and $|\mathbf{q}_0 + \nabla_{\perp} W/n'| < 1$.

For reference image $F = (X_0, Y_0, L)$, let $R_0 = \sqrt{L^2 + |\boldsymbol{\xi} - F_{\perp}|^2}$. Then

$$\mathbf{q}_0 = \frac{F_{\perp} - \boldsymbol{\xi}}{R_0}, \quad d_{0z} = \frac{L}{R_0}, \quad s(\mathbf{q}_0) = \frac{F_{\perp} - \boldsymbol{\xi}}{L}.$$

Consequently $\boldsymbol{\xi} + L s(\mathbf{q}_0) = F_{\perp}$ exactly, providing a direct reference-geometry check.

6.5 Paraxial propagation, the sign, and normalized coordinates

If both directions are close to $+z$, then $d_z \simeq d_{0z} \simeq 1$ and $s(\mathbf{q}) \simeq \mathbf{q}$. Equation (6.4) becomes textbook Eq. (4.4):

$$\Delta \boldsymbol{\xi} \simeq \frac{L}{n'} \nabla_{\perp} W. \quad (6.5)$$

The sign is positive because both errors are actual minus ideal. For example, $W_x > 0$ adds a positive x direction component and, in this paraxial expression, moves the intercept toward positive x . If OPD is defined as ideal minus actual, the right-hand side changes sign.

Using (1.3),

$$\Delta \boldsymbol{\xi} \simeq \frac{L}{n'a} \nabla_{u,v} \widetilde{W}. \quad (6.6)$$

For the textbook's $W = 20 \text{ nm } (x/a)$ with $a = 5 \text{ mm}$, $L = 100 \text{ mm}$, and $n' = 1$,

$$W_x = \frac{2 \times 10^{-5} \text{ mm}}{5 \text{ mm}} = 4 \times 10^{-6}.$$

This is the transverse direction error, approximately the angular error in radians here. Multiplication by L gives

$$\Delta x = (100 \text{ mm})(4 \times 10^{-6}) = 0.0004 \text{ mm} = 0.4 \mu\text{m}, \quad \Delta y = 0.$$

The optical-path derivative is dimensionless; the final multiplication by L produces a length.

6.6 Finite-angle linearization: derive every entry of the Jacobian

We now make a distinct approximation: $\delta \mathbf{q}$ is small, but $\mathbf{q}_0$ may be large. Define the finite slope difference $\Delta \mathbf{s} = \mathbf{s}(\mathbf{q}_0 + \delta \mathbf{q}) - \mathbf{s}(\mathbf{q}_0)$. A Taylor expansion gives

$$\Delta \mathbf{s} = J(\mathbf{q}_0) \delta \mathbf{q} + O(|\delta \mathbf{q}|^2). \quad (6.7)$$

The linear variation alone will be denoted $\delta \mathbf{s}$. The Jacobian is $J_{ij} = \partial s_i / \partial q_j$.

From $s_i = q_i(1 - |\mathbf{q}|^2)^{-1/2}$, the product rule gives

$$\frac{\partial s_i}{\partial q_j} = \delta_{ij}(1 - |\mathbf{q}|^2)^{-1/2} + q_i \frac{\partial}{\partial q_j} (1 - |\mathbf{q}|^2)^{-1/2}.$$

The remaining derivative is

$$\frac{\partial}{\partial q_j} (1 - |\mathbf{q}|^2)^{-1/2} = -\frac{1}{2} (1 - |\mathbf{q}|^2)^{-3/2} (-2q_j) = \frac{q_j}{(1 - |\mathbf{q}|^2)^{3/2}}.$$

Thus

$$J_{ij} = \frac{\delta_{ij}}{\sqrt{1 - |\mathbf{q}|^2}} + \frac{q_i q_j}{(1 - |\mathbf{q}|^2)^{3/2}}. \quad (6.8)$$

In matrix form,

$$J(\mathbf{q}) = \frac{I}{\sqrt{1 - |\mathbf{q}|^2}} + \frac{\mathbf{q} \mathbf{q}^\top}{(1 - |\mathbf{q}|^2)^{3/2}}, \quad (6.9)$$

where the outer product is

$$\mathbf{q} \mathbf{q}^\top = \begin{pmatrix} q_x^2 & q_x q_y \\ q_x q_y & q_y^2 \end{pmatrix}.$$

This is not the scalar dot product. Equivalently,

$$J = \frac{1}{(1 - q_x^2 - q_y^2)^{3/2}} \begin{pmatrix} 1 - q_y^2 & q_x q_y \\ q_x q_y & 1 - q_x^2 \end{pmatrix}.$$

Substitution of (6.2) gives the final matrix equation in Section 4.4:

$$\delta \mathbf{s} = \left[\frac{I}{\sqrt{1 - q_0^2}} + \frac{\mathbf{q}_0 \mathbf{q}_0^\top}{(1 - q_0^2)^{3/2}} \right] \frac{\nabla_\perp W}{n'}, \quad q_0^2 = |\mathbf{q}_0|^2. \quad (6.10)$$

The image error is $L \Delta \mathbf{s}$, approximately $L \delta \mathbf{s}$.

6.7 Physical interpretation of the finite-angle factors

Rotate the transverse axes so $\mathbf{q}_0 = (\sin \alpha_0, 0)$. Then

$$J(\mathbf{q}_0) = \begin{pmatrix} \sec^3 \alpha_0 & 0 \\ 0 & \sec \alpha_0 \end{pmatrix}.$$

For a perturbation parallel to the transverse ideal direction, $\delta s_{\parallel} = \sec^3 \alpha_0 \delta q_{\parallel}$. For a perpendicular perturbation, $\delta s_{\perp} = \sec \alpha_0 \delta q_{\perp}$. The parallel result can also be obtained directly:

$$\frac{d(\tan \alpha)}{d(\sin \alpha)} = \frac{\sec^2 \alpha}{\cos \alpha} = \sec^3 \alpha.$$

A steep ray needs more transverse travel per unit axial advance. This amplifies the change in slope caused by a given direction-cosine error. As $\alpha_0 \rightarrow 0$, both factors approach one and $J \rightarrow I$.

Keep distinct. Small wavefront amplitude, small direction perturbation, and small aperture angle are different conditions. A small-amplitude rapidly varying W can have a large gradient. Near $|\mathbf{q}_0| = 1$, the Jacobian grows strongly, and the linearized approximation is not uniform.

6.8 Why differentiating the next wavefront term is not sufficient

The small- q_0 expansion of (6.9) is

$$J = I + \frac{1}{2} q_0^2 I + \mathbf{q}_0 \mathbf{q}_0^T + O(q_0^4).$$

If the Gaussian reference is chosen so W starts at fourth total degree, $W = W_4 + W_6 + \dots$, then ∇W_4 is third degree and ∇W_6 is fifth degree. The quadratic correction in J multiplies ∇W_4 to generate another fifth-degree ray contribution:

$$\text{fifth-degree ray terms include } \nabla W_6 + J_2 \nabla W_4.$$

Here polynomial degree and perturbation order in $\delta \mathbf{q}$ are different classifications.

For an explicit on-axis meridional example, take $W = A_4 x^4 + A_6 x^6 + \dots$ on a plane a distance L before the focus. Since $q_0 = -x/\sqrt{L^2 + x^2}$,

$$J = (1 - q_0^2)^{-3/2} = \left(1 + \frac{x^2}{L^2}\right)^{3/2} = 1 + \frac{3x^2}{2L^2} + \dots$$

Multiplying by $W'(x) = 4A_4 x^3 + 6A_6 x^5 + \dots$ gives

$$\Delta x = \frac{4LA_4}{n'} x^3 + \left(\frac{6LA_6}{n'} + \frac{6A_4}{n'L}\right) x^5 + \text{higher powers.} \quad (6.11)$$

The second part of the fifth-degree coefficient comes from converting direction cosine to slope. Nonlinear terms in δq enter beyond fifth degree in this example.

6.9 Pupil mapping and a curved evaluation surface

Suppose a ray label η is related to physical output-plane position by $x = \eta + \kappa \eta^3 + \dots$. A quartic wavefront transforms as

$$A_4 x^4 = A_4 \eta^4 + 4\kappa A_4 \eta^6 + \dots$$

Its gradient must also be transformed: $dW/dx = (dW/d\eta)/(dx/d\eta)$. In two dimensions, if $\boldsymbol{\xi} = \mathbf{f}(\boldsymbol{\eta})$ has nonsingular Jacobian $M = \partial\boldsymbol{\xi}/\partial\boldsymbol{\eta}$,

$$\nabla_{\boldsymbol{\eta}} \widetilde{W} = M^T \nabla_{\perp} W, \quad \nabla_{\perp} W = M^{-T} \nabla_{\boldsymbol{\eta}} \widetilde{W}.$$

For a curved surface $z = h(x, y)$, define $S_h(x, y) = S(x, y, h(x, y))$. The chain rule gives

$$\partial_x S_h = S_x + S_z h_x = n'(d_x + d_z h_x), \quad \nabla_{x,y} S_h = n'(\mathbf{q} + d_z \nabla h).$$

Consequently

$$\nabla W_h = n'[\delta\mathbf{q} + (d_z - d_{0z})\nabla h]. \quad (6.12)$$

Propagation to $z = L$ now uses the axial distance $L - h(x, y)$:

$$\Delta\boldsymbol{\xi} = [L - h(x, y)][\mathbf{s}(\mathbf{q}) - \mathbf{s}(\mathbf{q}_0)]. \quad (6.13)$$

These formulas explain the last warning in Section 4.4. A derivative with respect to mirror footprint is not automatically the physical transverse gradient on a flat exit plane.

6.10 A compact interpretation guide

Quantity or statement	Physical meaning
S varies across the pupil	The actual wave has spatially varying phase; this can be perfectly ideal.
W varies across the pupil	The phase differs spatially from the chosen ideal reference.
$\nabla_{\perp} W/n'$	Transverse direction-cosine error at the same output point.
$\Delta\mathbf{s}$	Difference of transverse travel per unit axial travel. It is not generally an angular difference.
$\Delta\boldsymbol{\xi} = L\Delta\mathbf{s}$	Image-plane ray-intercept displacement for fixed planes.
Constant gradient	A uniform direction-cosine offset; a uniform image shift in the paraxial limit.
Gradient varying across pupil	Different ray displacements; generally a geometrical blur.
An off-centre pupil ray	A different ray from the same object point, not automatically an aberrated ray.

The ray calculation describes intersections, not the full diffraction intensity. Design, fabrication, alignment, and reference-plane choice can all contribute to a given W ; Section 4.4 maps that given error to ray positions without identifying its cause.

7 Worked example: an exact paraboloidal mirror

7.1 Define the physical system and the two meanings of small angle

An ideal paraboloid illuminated by an axial plane wave has zero ray-slope error relative to its exact axial focus at every usable aperture angle. Its nonzero reflected slope must not be mistaken for an error. We first prove that baseline, then tilt the incident plane wave to create a nontrivial example.

There are two independent angles:

1. the *field angle* β of the incident beam relative to the incoming axial direction $-z$;
2. the *aperture angle* of an outgoing ray relative to $+z$, which can be large on a fast mirror even when β is tiny.

The exact calculation below permits either angle to be non-small, subject to the indicated forward-ray conditions.

Let the mirror be

$$z = h(x_s, y_s) = \frac{x_s^2 + y_s^2}{4f}, \quad f > 0, \quad (7.1)$$

with vertex at the origin and axial focus $(0, 0, f)$. Subscript s labels the surface footprint, not the later flat-plane pupil coordinate. Use homogeneous index n on both paths and ideal specular reflection with spatially constant coating phase. Set

$$u_s = \frac{x_s}{2f}, \quad v_s = \frac{y_s}{2f}, \quad t = u_s^2 + v_s^2, \quad h = ft. \quad (7.2)$$

These u_s, v_s use the scale $2f$ and are not the earlier unit-disk coordinates $x/a, y/a$.

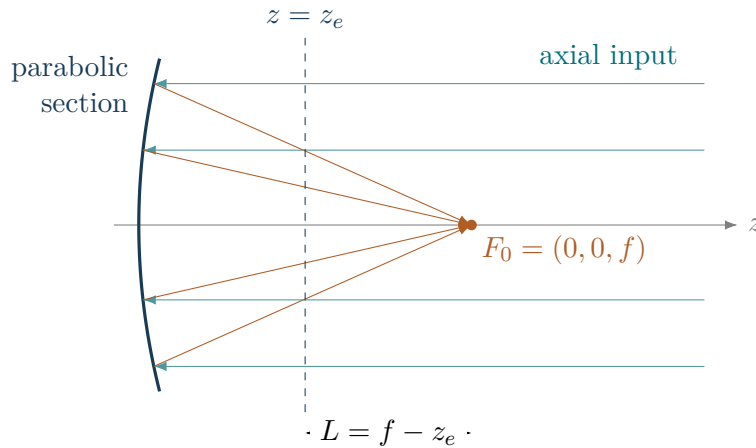


Figure 1: Meridional geometry of the axial baseline. Horizontal distance is z and vertical distance is a transverse coordinate. The dashed plane is a common downstream evaluation plane. The drawing is schematic.

7.2 Surface normal and exact axial reflection

The surface equation $G = z - (x_s^2 + y_s^2)/(4f) = 0$ has normal

$$\nabla G = (-u_s, -v_s, 1), \quad \mathbf{N} = \frac{(-u_s, -v_s, 1)}{\sqrt{1+t}}.$$

For $\mathbf{d}_{\text{in}} = (0, 0, -1)$, $\mathbf{d}_{\text{in}} \cdot \mathbf{N} = -1/\sqrt{1+t}$. Using (3.2),

$$\boxed{\mathbf{d}_a = \frac{(-2u_s, -2v_s, 1-t)}{1+t}}. \quad (7.3)$$

Its norm is one because $4t + (1-t)^2 = (1+t)^2$. For $t < 1$, it propagates along positive z . Its slope is

$$\mathbf{s}_a = -\frac{2(u_s, v_s)}{1-t} = -\frac{(x_s, y_s)}{f-h}. \quad (7.4)$$

Propagation from the mirror point to $z = f$ gives $(x_s, y_s) + (f-h)\mathbf{s}_a = (0, 0)$ exactly. Thus the ideal ray toward the axial focus has exactly the same slope:

$$\boxed{\Delta \mathbf{s} = \mathbf{0} \quad \text{for axial plane-wave input.}} \quad (7.5)$$

The actual slope itself can be large. In a meridional section, $s_a = -2u_s/(1-u_s^2)$, while $s_a \simeq -2u_s = -x_s/f$ for $|u_s| \ll 1$. The exact angle is $-2 \arctan u_s$. Error remains zero in both descriptions when each uses a consistent ideal reference.

7.3 Independent optical-path proof of zero axial aberration

Since $r_s^2 = 4fh$,

$$|F_0 - Q_s|^2 = r_s^2 + (f-h)^2 = (f+h)^2.$$

For $f+h > 0$, the distance is exactly $f+h$. An incoming equal-phase plane at $z = z_{\text{in}}$ contributes distance $z_{\text{in}} - h$ before reflection. The total optical path to focus is

$$n[(z_{\text{in}} - h) + (f+h)] = n(z_{\text{in}} + f),$$

independent of footprint. This is the exact constant-phase condition. There is no residual spherical aberration merely because h or the reflected angle is large.

7.4 Tilt the incident wave: exact reflected direction

Now choose the incident unit direction

$$\mathbf{d}_{\text{in}} = (\sin \beta, 0, -\cos \beta), \quad |\beta| < \pi/2.$$

Its dot product with the normal is

$$\mathbf{d}_{\text{in}} \cdot \mathbf{N} = -\frac{u_s \sin \beta + \cos \beta}{\sqrt{1+t}}.$$

Substitute into the reflection law and collect the three components:

$$\boxed{\mathbf{d}_a = \frac{1}{1+t} \begin{pmatrix} (1-u_s^2+v_s^2) \sin \beta - 2u_s \cos \beta \\ -2v_s(\cos \beta + u_s \sin \beta) \\ (1-t) \cos \beta + 2u_s \sin \beta \end{pmatrix}}. \quad (7.6)$$

No small-angle approximation has been made. Continue with footprints for which $d_{a,z} > 0$. Define

$$T = \tan \beta, \quad A = 1-t.$$

Dividing the transverse components by the longitudinal component gives

$$\mathbf{s}_a = \frac{(T(1-u_s^2+v_s^2) - 2u_s, -2v_s(1+u_s T))}{A + 2u_s T}. \quad (7.7)$$

This is an exact slope, not a direction cosine and not yet an error.

In the meridional plane $y_s = 0$, let $\gamma = \arctan u_s$. Since $\sin(2\gamma) = 2u_s/(1 + u_s^2)$ and $\cos(2\gamma) = (1 - u_s^2)/(1 + u_s^2)$,

$$d_{a,x} = \sin(\beta - 2\gamma), \quad d_{a,z} = \cos(\beta - 2\gamma).$$

Thus the signed ray angle and its slope are

$$\alpha_a = \beta - 2 \arctan u_s, \quad s_a = \tan \alpha_a \quad (7.8)$$

on the forward branch. This independently checks the vector calculation.

7.5 Specify the reference image before defining an error

For the tilted input, choose

$$F_\beta = (f \tan \beta, 0, f). \quad (7.9)$$

The vertex ray reflects in direction $(\sin \beta, 0, \cos \beta)$ and reaches this point on $z = f$. Therefore this choice follows the vertex ray and removes its overall lateral displacement from the definition of error. For a stop at the mirror, this is a natural chief-ray reference. At small field, its image height is $f\beta$ to first order. At finite field, (7.9) is an explicit reference convention; it is not a claim that every ray focuses there.

The ideal straight line from the same mirror point to F_β has slope

$$\mathbf{s}_{0,s} = \frac{(fT - x_s, -y_s)}{f - h} = \frac{(T - 2u_s, -2v_s)}{A}. \quad (7.10)$$

We first calculate a slope difference using this common *surface* point, denoted $\Delta \mathbf{s}_s = \mathbf{s}_a - \mathbf{s}_{0,s}$. We subsequently transfer the comparison to the common flat plane required by textbook Section 4.4.

For the x component, subtract the fractions:

$$\Delta s_{x,s} = \frac{A[T(1 - u_s^2 + v_s^2) - 2u_s] - (T - 2u_s)(A + 2u_sT)}{A(A + 2u_sT)} \quad (7.11)$$

$$= \frac{AT(-u_s^2 + v_s^2) - 2u_sT^2 + 4u_s^2T}{A(A + 2u_sT)}. \quad (7.12)$$

Use $4 - A = 3 + t$ to obtain

$$\Delta s_{x,s} = \frac{T[(3 + t)u_s^2 + (1 - t)v_s^2 - 2u_sT]}{(1 - t)(1 - t + 2u_sT)}. \quad (7.13)$$

Similarly,

$$\Delta s_{y,s} = \frac{-2v_s(1 + u_sT)A + 2v_s(A + 2u_sT)}{A(A + 2u_sT)} \quad (7.14)$$

$$= \frac{2u_s v_s T(1 + t)}{(1 - t)(1 - t + 2u_sT)}. \quad (7.15)$$

Both vanish when $\beta = 0$, as required by the axial result. They also vanish at the vertex, consistently with the chosen reference.

7.6 Put both rays on a common flat output plane

Choose a plane $z = z_e$ beyond the mirror cap under consideration and before $z = f$. Let

$$L = f - z_e > 0, \quad h \leq z_e.$$

Assume a locally single-valued ray mapping at the chosen plane. The actual reflected ray reaches the transverse position

$$\boldsymbol{\xi}_e = (x_e, y_e) = (x_s, y_s) + (z_e - h)\mathbf{s}_a. \quad (7.16)$$

At this *same* point, the ideal ray toward F_β has

$$\mathbf{s}_{0,e} = \frac{F_{\beta,\perp} - \boldsymbol{\xi}_e}{L}, \quad (7.17)$$

$$R_0 = \sqrt{L^2 + |F_{\beta,\perp} - \boldsymbol{\xi}_e|^2}, \quad (7.18)$$

$$\mathbf{q}_{0,e} = \frac{F_{\beta,\perp} - \boldsymbol{\xi}_e}{R_0} = \frac{\mathbf{s}_{0,e}}{\sqrt{1 + |\mathbf{s}_{0,e}|^2}}. \quad (7.19)$$

The actual direction remains constant during free propagation, so $\mathbf{q}_a = (d_{a,x}, d_{a,y})$ from (7.6). The exact error used in Section 4.4 is now

$$\Delta \mathbf{s}_e = \mathbf{s}(\mathbf{q}_a) - \mathbf{s}(\mathbf{q}_{0,e}) = \mathbf{s}_a - \mathbf{s}_{0,e}. \quad (7.20)$$

To relate it to the surface comparison, multiply by L and substitute (7.16):

$$L\Delta \mathbf{s}_e = L\mathbf{s}_a - F_{\beta,\perp} + \boldsymbol{\xi}_e \quad (7.21)$$

$$= (f - h)\mathbf{s}_a + (x_s, y_s) - F_{\beta,\perp} \quad (7.22)$$

$$= (f - h)(\mathbf{s}_a - \mathbf{s}_{0,s}). \quad (7.23)$$

Consequently,

$$\Delta \mathbf{s}_e = \frac{f - h}{L} \Delta \mathbf{s}_s, \quad \Delta \boldsymbol{\xi}_{\text{image}} = L\Delta \mathbf{s}_e = (f - h)\Delta \mathbf{s}_s. \quad (7.24)$$

This also follows by tracing the actual ray all the way to $z = f$ and subtracting $F_{\beta,\perp}$.

Interpretation. The actual ray slope does not change between these planes. The paired ideal ray changes because it must begin at the new common point and still aim at the reference image. That is why the slope *error* at a surface point and at a flat-plane point need not be numerically identical.

7.7 Recover the same result from an eikonal on that plane

The incident plane wave has eikonal $n(\sin \beta x - \cos \beta z)$ plus a constant. At a mirror point, the reflected wave inherits that phase, apart from the assumed spatially constant coating phase. The reflected path from $z = h$ to $z = z_e$ has geometrical length $(z_e - h)/d_{a,z}$. Therefore

$$S_a(\boldsymbol{\xi}_e) = n \left[\sin \beta x_s - \cos \beta h + \frac{z_e - h}{d_{a,z}} \right] + C_a. \quad (7.25)$$

Here x_s, y_s label the ray that arrives at $\boldsymbol{\xi}_e$; this is a parametric representation of an eikonal on the *flat* plane. The ideal converging wave is $S_0 = C_0 - nR_0$. One convenient piston choice gives

$$W_e(\boldsymbol{\xi}_e) = n \left[\sin \beta x_s - \cos \beta h + \frac{z_e - h}{d_{a,z}} + R_0 - \frac{f}{\cos \beta} \right]. \quad (7.26)$$

At the vertex-labelled ray, $\boldsymbol{\xi}_e = (z_e \tan \beta, 0)$ and $R_0 = L / \cos \beta$; hence $W_e = 0$ there. This piston choice does not affect any ray error.

The eikonal-gradient law gives

$$\frac{\nabla_{\perp} W_e}{n} = \mathbf{q}_a - \mathbf{q}_{0,e} \equiv \delta \mathbf{q}. \quad (7.27)$$

This is the direct bridge to Section 4.4. If differentiating the parametric formula explicitly, the coordinate mapping cannot be skipped. In a meridional section,

$$\frac{dW_e}{dx_e} = \frac{dW_e/dx_s}{dx_e/dx_s}. \quad (7.28)$$

In two dimensions, the corresponding inverse-transpose Jacobian rule was derived in Section 6. Differentiating with respect to a curved-surface coordinate and calling it a flat-pupil gradient would yield a different, generally incorrect, quantity.

7.8 Small field versus small aperture: take the limits explicitly

First let $\beta \rightarrow 0$ at a fixed surface footprint. Since $T = \beta + O(\beta^3)$, (7.13)–(7.15) give

$$\Delta s_{x,s} = \beta \frac{(3+t)u_s^2 + (1-t)v_s^2}{(1-t)^2} + O(\beta^2), \quad (7.29)$$

$$\Delta s_{y,s} = \beta \frac{2u_s v_s (1+t)}{(1-t)^2} + O(\beta^2). \quad (7.30)$$

These formulas are first order in field angle but retain finite aperture. They need not reduce to the paraxial direction-to-slope rule.

Next suppose $|u_s|, |v_s| \ll 1$ as well. Keeping the leading pupil degree in the coefficient of β gives

$$\Delta \mathbf{s}_s \simeq \beta(3u_s^2 + v_s^2, 2u_s v_s) = \frac{\beta}{4f^2}(3x_s^2 + y_s^2, 2x_s y_s). \quad (7.31)$$

Multiplying by $f - h \simeq f$ gives the leading transverse image error

$$\Delta \boldsymbol{\xi}_{\text{image}} \simeq \frac{\beta}{4f}(3x_s^2 + y_s^2, 2x_s y_s). \quad (7.32)$$

The pupil-quadratic, field-linear pattern is the leading coma term of this mirror. This statement concerns only the first-order dependence on β . Terms proportional to $\beta^2 u_s$, for example, were discarded earlier; the formula is not a complete expansion in all pupil and field variables through total degree three.

In the meridional plane, the finite-aperture, small-field result at the flat plane is particularly simple:

$$\Delta s_e = \frac{f-h}{L} \beta \frac{u_s^2(3+u_s^2)}{(1-u_s^2)^2} + O(\beta^2). \quad (7.33)$$

It grows strongly as the outgoing rays become steep. The exact expressions should be used when the discarded field terms are appreciable.

7.9 Three ways to calculate the flat-plane slope error

For a meridional ray, put $q_a = d_{a,x}$, $q_0 = q_{0,e,x}$, and $\delta q = q_a - q_0 = W'_e/n$. The three results to compare are

$$\Delta s_{\text{exact}} = \frac{q_a}{\sqrt{1 - q_a^2}} - \frac{q_0}{\sqrt{1 - q_0^2}}, \quad (7.34)$$

$$\Delta s_{\text{parax}} \simeq \delta q, \quad (7.35)$$

$$\Delta s_{\text{linear}} \simeq \frac{\delta q}{(1 - q_0^2)^{3/2}}. \quad (7.36)$$

The second line assumes small outgoing angles. The third line permits a large ideal outgoing angle but linearizes in the direction error δq . It is the longitudinal-to- $\mathbf{q}_0$ eigenvalue of the two-dimensional Jacobian from Section 4.4, since all rays here lie in one plane.

In the numerical comparisons below, all three formulas use the same exact δq from the actual and reference rays. Thus the difference between them isolates the approximation in *converting direction error to slope error*; it does not mix in a truncated wavefront model.

7.10 Case A: small incident angle and small outgoing angle

Use millimetres throughout and take

$$f = 100 \text{ mm}, \quad z_e = 50 \text{ mm}, \quad L = 50 \text{ mm}, \quad n = 1, \quad x_s = 10 \text{ mm}, \quad y_s = 0, \quad \beta = 10^{-4} \text{ rad}.$$

The mirror and field parameters are

$$u_s = 0.05, \quad t = 0.0025, \quad h = 0.25 \text{ mm}, \quad T = 0.0001000000003333.$$

1. **Reflect the actual ray.** Equation (7.6) gives

$$q_a = -0.099651121696, \quad d_{a,z} = 0.995022438915, \quad s_a = -0.100149622560.$$

2. **Locate it on the evaluation plane.**

$$x_e = 10 + (50 - 0.25)(-0.100149622560) = 5.017556277648 \text{ mm}.$$

The reference image height is $X_\beta = fT = 0.0100000000333 \text{ mm}$.

3. **Construct the ideal ray from this same point.**

$$s_0 = \frac{X_\beta - x_e}{50} = -0.100151125552, \quad q_0 = \frac{s_0}{\sqrt{1 + s_0^2}} = -0.099652602356.$$

Its aperture angle has magnitude about 5.719° .

4. **Subtract directions and slopes separately.**

$$\begin{aligned} \delta q &= 1.480659945 \times 10^{-6}, \\ \Delta s_{\text{exact}} &= 1.502992456 \times 10^{-6}, \\ \Delta s_{\text{parax}} &= 1.480659945 \times 10^{-6}, \\ \Delta s_{\text{linear}} &= 1.015083036 \delta q = 1.502992792 \times 10^{-6}. \end{aligned}$$

5. **Convert slope error to image displacement.**

$$\Delta x_{\text{image}} = 50 \Delta s_{\text{exact}} = 7.514962281 \times 10^{-5} \text{ mm} = 0.0751496 \mu\text{m}.$$

The paraxial result underestimates the slope error by about 1.49%. The finite-angle linearization is essentially exact at the displayed precision because the direction perturbation is very small. For a separate small-field check, (7.33) gives $1.505012531 \times 10^{-6}$; also keeping only the leading pupil degree gives $(f-h)3\beta u_s^2/L = 1.49625 \times 10^{-6}$. These last two values additionally approximate the mirror's field dependence, unlike (7.35).

The OPD from (7.26) is $W_e \simeq 2.48883 \text{ nm}$ at this ray. Its local derivative, rather than this single OPD value, sets δq . The derivative with respect to x_e is $1.480659945 \times 10^{-6}$ for $n = 1$.

7.11 Case B: small incident angle but a large outgoing angle

Keep the same f, z_e, L, n, β , but choose

$$x_s = 100 \text{ mm}, \quad y_s = 0, \quad u_s = 0.5, \quad t = 0.25, \quad h = 25 \text{ mm}.$$

This ray lies at the edge of a 200 mm diameter cap if that cap is used. The small illumination angle remains 10^{-4} radian, yet the outgoing ray is steep.

1. Actual reflection.

$$q_a = -0.799939996000, \quad d_{a,z} = 0.600079997000, \quad s_a = -1.333055592587.$$

2. Common point on $z = 50 \text{ mm}$.

$$x_e = 100 + 25s_a = 66.673610185332 \text{ mm}.$$

3. Reference ray.

$$s_0 = \frac{0.0100000000333 - 66.673610185332}{50} = -1.333272203706,$$

$$q_0 = -0.799986795419, \quad |\alpha_0| = |\arctan s_0| \simeq 53.129^\circ.$$

4. Direction and exact slope differences.

$$\delta q = 4.679941922 \times 10^{-5}, \quad \Delta s_{\text{exact}} = 2.166111192 \times 10^{-4}.$$

5. Compare the approximations.

$$\begin{aligned} \Delta s_{\text{parax}} &= 4.679941922 \times 10^{-5}, \\ (1 - q_0^2)^{-3/2} &= 4.629222114, \\ \Delta s_{\text{linear}} &= 2.166449064 \times 10^{-4}. \end{aligned}$$

6. Image displacement.

$$\Delta x_{\text{image}} = 50 \Delta s_{\text{exact}} = 0.01083055596 \text{ mm} = 10.8306 \mu\text{m}.$$

The paraxial conversion is about 78.4% too small. The finite-angle linearization differs from the exact result by only 0.0156%. A tiny field angle is therefore insufficient to justify $s \simeq q$: the relevant reference aperture angle is large.

The first-order-in-field expression with exact aperture dependence, (7.33), yields $2.166666667 \times 10^{-4}$, close to the exact answer. Dropping that aperture dependence and using the leading pupil term instead gives only 1.125×10^{-4} .

7.12 Case C: finite field and finite aperture

Keep the Case B geometry, but increase the incident angle to $\beta = 10^\circ = \pi/18$ radian. The exact formulas still apply:

$$T = 0.176326980708, \quad X_\beta = 17.632698070846 \text{ mm.}$$

Stepwise evaluation gives

$$\begin{aligned} q_a &= -0.683657295810, & d_{a,z} &= 0.729803193941, \\ s_a &= -0.936769394113, & x_e &= 76.580765147179 \text{ mm}, \\ s_0 &= -1.178961341527, & q_0 &= -0.762614697904. \end{aligned}$$

The reference aperture angle is about 49.695° . Consequently,

$$\begin{aligned} \delta q &= 0.078957402094, \\ \Delta s_{\text{exact}} &= (-0.936769394113) - (-1.178961341527) = 0.242191947414, \\ \Delta s_{\text{parax}} &= 0.078957402094, \\ \Delta s_{\text{linear}} &= 3.694734029 \delta q = 0.291726600344, \\ \Delta x_{\text{image}} &= 50(0.242191947414) = 12.1095973707 \text{ mm.} \end{aligned}$$

Here the paraxial result is 67.4% too small and the finite-angle linearization is 20.5% too large. The second approximation now fails because the change in direction is no longer sufficiently small for a first-order Taylor expansion at q_0 . The exact slope function is needed.

Quantity	Case A	Case B	Case C
x_s (mm)	10	100	100
β (rad)	10^{-4}	10^{-4}	$\pi/18$
$ \alpha_0 $ (degrees)	5.719	53.129	49.695
Exact Δs_e	1.50299×10^{-6}	2.16611×10^{-4}	0.242192
Paraxial δq	1.48066×10^{-6}	4.67994×10^{-5}	0.0789574
Finite-angle linearization	1.50299×10^{-6}	2.16645×10^{-4}	0.291727
Exact image error (mm)	7.51496×10^{-5}	0.0108306	12.1096

Table 2: All slope differences use the same flat plane $z_e = 50$ mm and the reference image $(f \tan \beta, 0, f)$, with $f = 100$ mm. The linearization retains the exact reference aperture angle.

7.13 What this example establishes

The axial paraboloid has zero slope error, even for steep rays. Tilting its illumination introduces a footprint-dependent error. The wavefront and reflection calculations agree through $\nabla_\perp W/n = \delta \mathbf{q}$ on the common plane. Case A supports the paraxial conversion; Case B requires the finite-angle Jacobian; Case C requires the exact slope difference.

Keep distinct. A slope difference Δs is dimensionless, and it is approximately an angular error only in the paraxial limit. The actual angular error in a meridional plane is $\Delta \alpha = \arctan s_a - \arctan s_0$ on a consistent branch. The transverse ray error is the length $L \Delta s$. These three quantities answer different questions.